Comprehensive Project Report: Healthcare Clinical & Financial Analytics

1. Project Overview & Dataset

This project analyzes a synthetic but highly realistic healthcare dataset representing over 100,000 patients and 3 million clinical records. The goal was to identify operational bottlenecks, demystify hospital billing structures, and uncover clinical risk factors driving patient readmissions and mortality.

The Relational Database Schema:

- **Patients:** Demographics, lifestyle factors, and Charlson Comorbidity Index (100,000 rows).
- **Outcomes:** The central fact table tracking admissions, ICU stays, lengths of stay (LOS), charges, and mortality (11,001 rows).
- **Diagnoses:** Transactional records of primary and secondary diseases per visit (274k+ rows).
- **Medications:** Prescription adherence data (364k+ rows).
- **Lab Results:** Granular, row-level diagnostic test results (2.2M+ rows).

2. Steps Involved

1. **Data Extraction & Engineering (Python/Pandas):** Cleaned messy text columns, normalized pipe-delimited strings (secondary diagnoses) into relational bridging tables, and engineered new age-bucket features.
2. **Database Architecture (MySQL):** Designed a robust relational database schema and executed optimized, chunked data loads to securely import over 3 million rows without memory failures.
3. **Exploratory Data Analysis (SQL):** Conducted rigorous EDA using CTEs, complex aggregations, and multi-table joins to test clinical and financial hypotheses.
4. **Semantic Layer & Dimensional Modelling:** Created a "Star Schema" using SQL Views to pre-aggregate 2.2M lab records into patient-level summaries, protecting downstream performance.
5. **Business Intelligence (Power BI):** Designed a 2-page, interactive executive dashboard highlighting financial operations and clinical risks utilizing DAX measures and cross-filtering.

3. Key Technical Challenges & Solutions

- **The "Cartesian Smear" (Grain Mismatch):** Initial financial queries joining patient outcomes to transactional diagnoses caused a Cartesian product, uniformly duplicating the $18,000 hospital charge across all diseases. *Solution:* Audited row counts, identified the many-to-many relationship trap, and isolated financial analysis to strict 1-to-1 patient outcome queries.
- **Referential Integrity Failures (The Blank Slicer):** In Power BI, joining fact tables directly to other fact tables generated "ghost rows" (Blank) in filters due to mismatched patient populations (100k vs 11k). *Solution:* Re-architected the data model into a strict Star Schema, routing all filters through a central dimension table (dim_patients).
- **Visualizing Extreme Anomalies:** Mortality rates (1.3%) and readmission spikes were lost when averaged out or placed in 100% stacked bar charts. *Solution:* Utilized standard column charts

for mortality and implemented “Binning" (grouping lab failures into clusters of 15) to clearly visualize the exponential risk correlation.

4. Analysis & Business Insights

Through rigorous scenario testing, the following truths were uncovered:

- **Financial Insight: DRG Flat-Rate Billing:** The data definitively proved the hospital utilizes Diagnosis-Related Group (DRG) billing rather than Fee-For-Service. Whether a patient required intensive ICU care or a standard ward bed, average charges hovered strictly around $18,000 based purely on their DRG code category.
- **Socioeconomic Insight: The Uninsured Penalty:** Uninsured patients received the lowest overall bills (due to massive self-pay discounts/write-offs), but they suffered the highest mortality rate (1.53%) and the longest average length of stay (5.30 days). This indicates a delay in seeking medical care due to cost fears, resulting in higher patient acuity upon eventual admission.
- **Clinical Insight: The 26% Readmission Spike:** Extensive scenario testing proved that Age alone is a baseline gravity for readmissions. However, when combining the Senior age demographic with high clinical complexity (patients experiencing >40 abnormal lab results during their stay), the 30-day readmission rate exponentially spiked to **26.32%**.
- **Operational Insight: Defensive Medicine / Over-testing:** Querying lab result volumes against primary diagnoses revealed that for patients presenting with Anxiety or Coronary Artery Disease, over 83% to 89% of ordered lab panels returned perfectly normal results, indicating significant potential for over-testing.

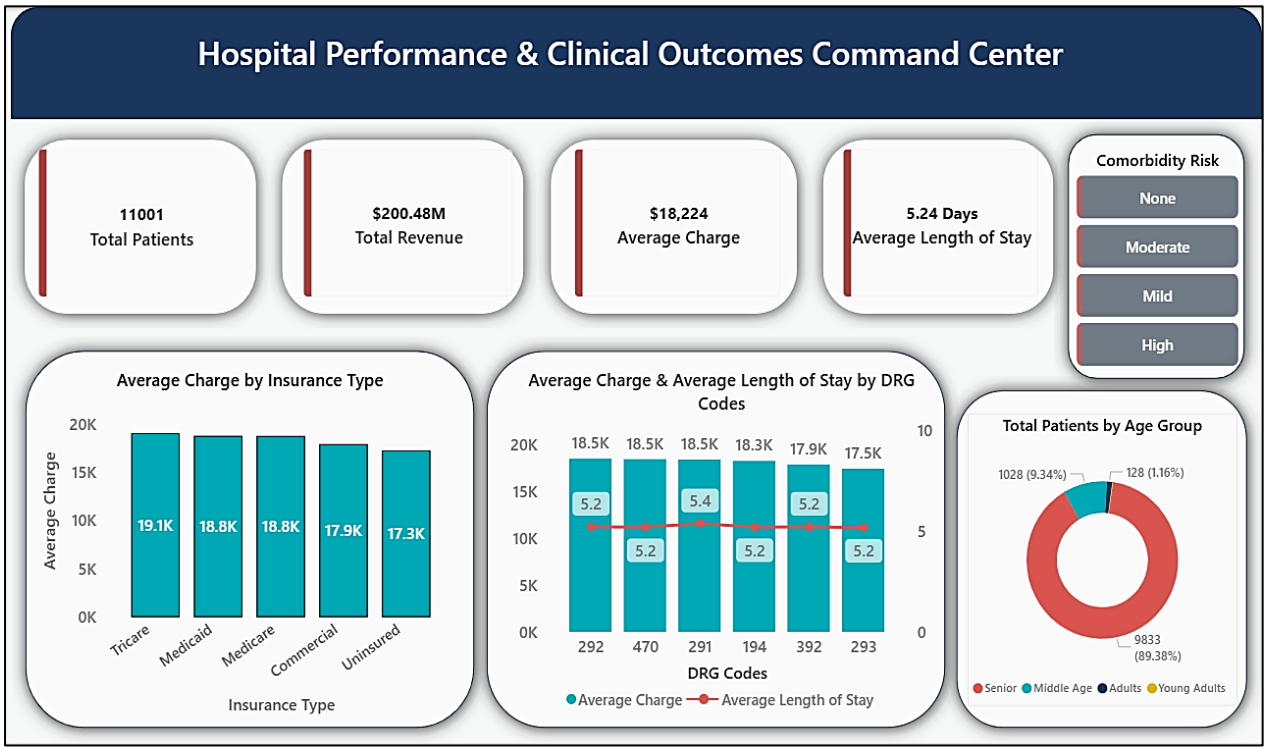
disease	total_test_vol	normal_results	rate_normal_results
Hyperlipidemia	668	422	63.17
Hypertension	641	464	72.39
Obesity	529	405	76.56
Chronic Kidney Disease	284	182	64.08
Anxiety	187	157	83.96
Type2 Diabetes	161	105	65.22
Osteoarthritis	160	120	75.00
Coronary Artery Disease	114	102	89.47
Asthma	112	89	79.46
Copd	97	68	70.10
Depression	94	67	71.28
Heart Failure	80	63	78.75
Hypothyroidism	62	45	72.58
Atrial Fibrillation	28	22	78.57

5. Strategic Recommendations

1. **Implement Aggressive Post-Discharge Protocols:** The hospital must auto-flag Senior patients discharging with high volumes of abnormal lab results for mandatory 48-hour telehealth follow-ups to mitigate the 26% readmission risk.
2. **Audit Diagnostic Order Sets:** Medical directors should review automated lab-order sets for patients presenting with Anxiety and CAD to reduce unnecessary needle-sticks, save clinical hours, and cut operational waste on tests that return normal 89% of the time.
3. **Community Health Outreach:** The high mortality and prolonged LOS for uninsured patients strains emergency resources. Implementing early-intervention community health clinics or transparent financial-assistance marketing could encourage uninsured patients to seek care before their conditions become critical.

6. Dashboards

- **Financial & Operational Overview:**



- **Clinical Risk & Readmission Dynamics**

